

圖片手指放大縮小、平移

// 控制圖片位移與縮放的矩陣

```
private val imageMatrix = Matrix()
```

// 縮放與平移的狀態變數

```
private var scaleFactor = 1f
```

```
imageView = findViewById(R.id.imageView)
```

```
val scaleImageView: ImageView = findViewById(R.id.scaleImageView)
```

```
scaleImageView.scaleType = ImageView.ScaleType.MATRIX
```

// 圖像放大縮小

```
val scaleDetector = ScaleGestureDetector(this, object :  
ScaleGestureDetector.SimpleOnScaleGestureListener() {  
    override fun onScale(detector: ScaleGestureDetector): Boolean {  
        val scale = detector.scaleFactor  
        val nextScale = scaleFactor * scale  
  
        if (nextScale in 1f..5f) {  
            scaleFactor = nextScale  
            imageMatrix.postScale(scale, scale, detector.focusX,  
detector.focusY)  
            scaleImageView.imageMatrix = imageMatrix  
        }  
        return true  
    }  
})
```

// 圖像位移

```
val gestureDetector = GestureDetector(this, object :  
GestureDetector.SimpleOnGestureListener() {  
    override fun onScroll(  
        e1: MotionEvent?,  
        e2: MotionEvent,  
        distanceX: Float,  
        distanceY: Float  
    ): Boolean {  
        imageMatrix.postTranslate(-distanceX, -distanceY)  
        scaleImageView.imageMatrix = imageMatrix  
        return true  
    }  
})
```

// 設置接觸事件

```
scaleImageView.setOnTouchListener { _, event ->
    scaleDetector.onTouchEvent(event)
    gestureDetector.onTouchEvent(event)
    true
}
```



FOR CPTTM USE ONLY